

London School of Economics
EC400

Economics Department
September 2011

Part 2: Mathematics for Microeconomics Examination

Time: 1 hour

Write your name, your class group number, and the name of your class teacher on your script.

This part of your examination counts for 100 marks. Answer all 3 questions.

Explain your answers. Answers that are correct, but not explained do not get high marks.

PTO

1. (20 marks) Consider the following quadratic form:

$$Q(\mathbf{x}) = 2x_1^2 + 4x_3^2 + 4x_1x_2 - 2x_1x_3$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the leading principal submatrices of A .

- (b) (8 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *positive definite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is positive definite?
- (c) (8 marks) Is the quadratic form $Q(\mathbf{x})$ *positive definite*, *negative definite*, *positive semi-definite*, *negative semi-definite* or *indefinite*?
2. (40 marks) For some constant $k \in (0, 2)$ consider the function defined on $\mathbb{R}^2$:

$$f(x, y) = k(xy + x + y) - x^2 - y^2$$

- (a) (8 marks) Find the *critical points* of $f(x, y)$.
- (b) (8 marks) Are these critical points *local maximizers*, *local minimizers* or *neither*?
- (c) (8 marks) Are these critical points *global maximizers*, *global minimizers* or *neither*?
- (d) (8 marks) Is $f(x, y)$ *concave*, *convex*, *quasi-concave*, *quasi-convex* or *neither of the above*?
- (e) (8 marks) Denote the *global maximizer* of $f(x, y)$ by $(x(k), y(k))$. Use the envelope theorem to show how *maximum value function* responds to a marginal change in k . Also show how the *global maximizer* $(x(k), y(k))$ is affected by marginal changes in k .

3. (40 marks) For $I > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & F(x, y) = -e^{-xy} \\ \text{s.t.} \quad & x + 2y \leq I \\ & x \geq 0 \\ & y \geq 0 \end{aligned} \tag{1}$$

where e^z denotes the exponential function evaluated at z .

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Which conditions on the concavity/convexity of such functions would suffice to guarantee that the Lagrangian is concave?
- (b) (8 marks) Set up the constrained optimization problem using the Lagrangian function. Can we solve this optimization problem using the Kuhn-Tucker method? Are the Kuhn-Tucker constraint qualifications satisfied? Is it possible to solve the same optimization problem with the Lagrange method? Are the Lagrange constraint qualifications satisfied?
- (c) (8 marks) Using the Kuhn-Tucker approach, write down the necessary first order conditions for the solution of the constrained optimization problem.
- (d) (4 marks) Are these conditions also *sufficient* to characterize an optimum in this problem?
- (e) (4 marks) Solve the constrained optimization problem in terms of I .
- (f) (8 marks) Let $x^*(I)$ and $y^*(I)$ be the solution of this problem. Find the slope of such functions with respect to I . Moreover, find how the maximum value function is affected by a change in I .